Neural Networks Project — Educational Platform

Overall Objective

Develop an interactive educational platform that enables users to:

- Understand how neural networks work
- Experiment with different models
- Visualize the learning process
- Compare performance

Overall Project Architecture

1. User Interface (UI)

The platform should allow the user to:

- Load a dataset (CSV)
- Choose the type of problem:
    - Binary classification
    - Multi-class classification
    - Regression
- Select a model

2. Models to Implement

Level 1 — Fundamentals

- Historical perceptron
- Perceptron (with activation function)

Level 2 — Advanced Neural Networks

Multi-Layer Perceptron (MLP)

Implement:

- Forward propagation

Model Parameterization

The user must be able to choose:

- Number of hidden layers
- Number of neurons per layer

- Activation functions:
    - Sigmoid
    - ReLU
    - Tanh
    - Softmax (multi-class output)

Level 3 — Learning and Optimization

Cost functions

- MSE (regression)

- Binary Cross-Entropy

- Categorical Cross-Entropy

Optimization

- Gradient Descent

- (Bonus) Mini-batch Gradient Descent

Backpropagation

Level 4 — Regularization & Generalization

Implement and compare:

Techniques

- L2 Regularization
- Dropout (advanced bonus)
- Early stopping

Pedagogical objective

Understand:

- Overfitting
- Underfitting
- Bias vs Variance

Level 5 — Model Evaluation

Classification

- Accuracy
- Precision / Recall
- F1-score
- Confusion matrix

Regression

- MSE
- RMSE
- R^2 score

Level 6 — Visualization (VERY IMPORTANT)

The platform must display:

During training

- Loss vs Epochs
- Accuracy vs Epochs

Results

- Decision boundary (2D)
- Train vs Test comparison

This is what makes the project educational, not just technical.

Recommended Technical Architecture

Backend (Python)

Modules to develop:

- models/

  - perceptron.py
  - mlp.py
  - rbf.py

- training/

  - forward.py
  - backward.py
  - optimizer.py

- utils/

  - metrics.py
  - activation.py
  - loss.py

Frontend (depending on level)

- Simple option: Streamlit

- Advanced option: Flask + HTML/CSS/JS

Key Platform Features

Configuration

- Model selection
- Hyperparameters:
    - Learning rate
    - Epochs
    - Architecture

Data

- CSV upload
- Train / Test split

Execution

- “Train” button
- Real-time display

Mandatory Experiments

Experiment 1

Compare:

- Perceptron vs MLP

Experiment 2

Effect of the number of layers

Experiment 3

Overfitting vs regularization

Expected Deliverables

Code

- Well structured
- Documented

Report

Must include:

- Theoretical explanation
- Experimental results
- Critical analysis

Demo

- Platform presentation
- Use cases

Evaluation Criteria

Criterion	Weight
Correct implementation	30%
Theoretical understanding	20%
Experimentation	20%
Platform quality	20%
Innovation / bonus	10%

Possible Bonuses

- Comparison with sklearn

- Advanced interactive interface

- 3D visualization

"You are not just implementing neural networks — you are building a platform that teaches how they work."